VICTORY 2027

The Endocrine System

LECTURE-02 BIOLOGY

By : Upasana Ma'am



Topics to be Covered

1. Thyroid
2. Pancreas
3. Adrenal Gland
4.


HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上

Differences between Hormonal Control and Nervous Control

Hormonal control	Nervous control
1. Usually slow.	1. Immediate/Rapid.
2. Transmitted chemically through blood.	2. Transmitted electrochemically through the nerve fibres and chemically across synapses.
3. Affects different organs (widespread in body).	3. Affects only the particular muscles or the gland (local).
4. Effect is short term or long-lasting.	4. Effect only short-lived.
5. Can affect growth.	5. Cannot affect growth.
6. Can bring about specific chemical changes and regulates metabolism.	6. Does not influence chemical changes and cannot regulate metabolism.
7. Cannot be modified by learning from previous experience.	7. Can be modified to some extent by learning from previous experience.

B. Oversecretion (Hyperthyroidism) :

Excess of thyroxine secretion may also cause a kind of goitre called **exophthalmic goitre** (exo : outward, ophthalmos: eye).



- ✓ protruding eyes
- ✓ Goitre.

- A person having the problem of over - secretion shows:

- Protrusion of eyes,
- Formation of goitre in the neck,
- A marked increase in metabolic rate,
- Rapid heart-beat, and
- Shortness of breath.



Thyroxine

$AB \rightarrow A + B$

150 ✓

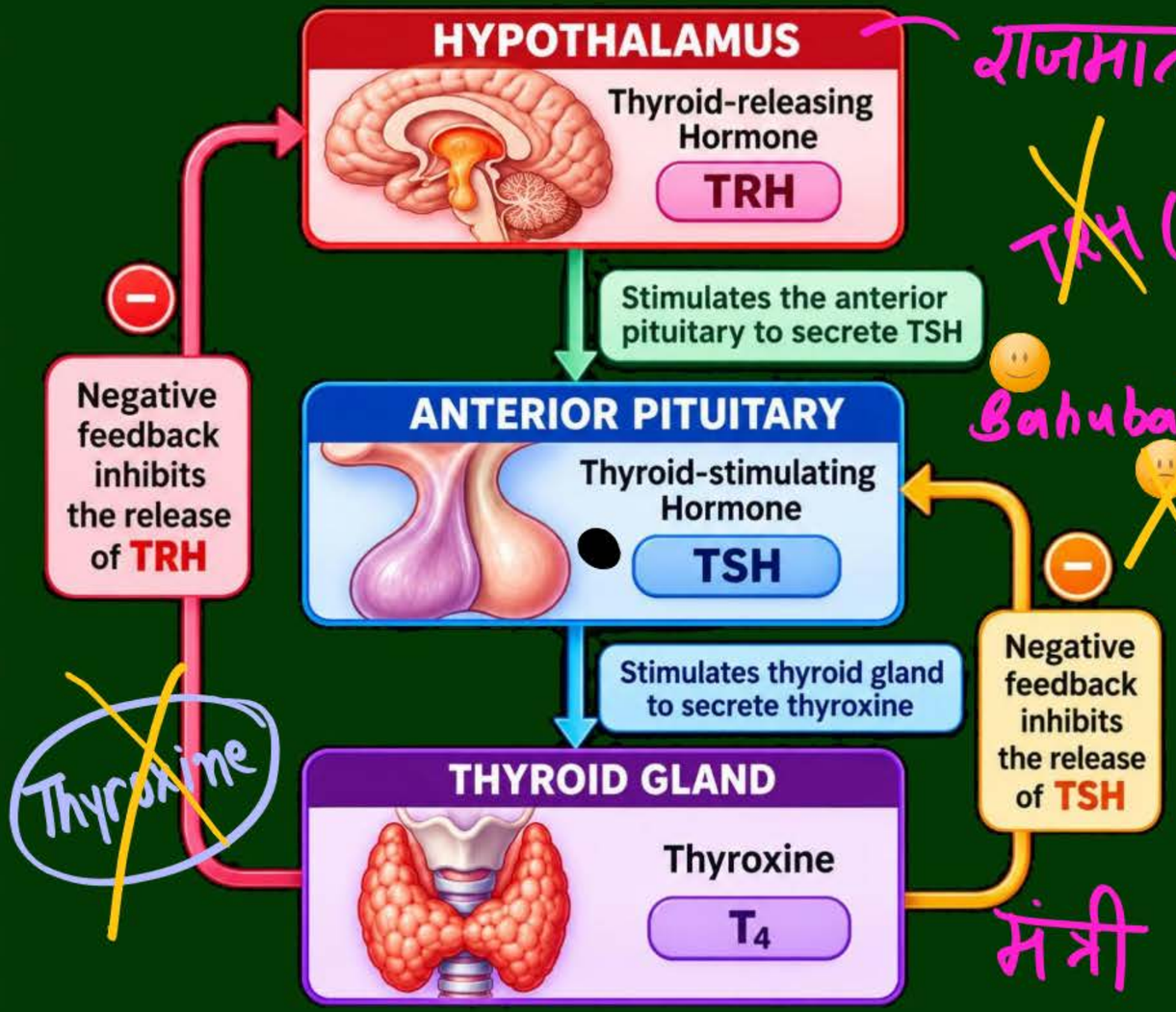
Control of Hormonal Secretions [Feedback Mechanism]

- [Commands like saying "Too much, slow down" or "Too little, speed up"] - This is the feedback mechanism.
- The body has a mechanism to maintain a normal state. Whenever there is a change in this state there are "messages" through the body systems to "increase" if there is a fall below the normal, or to "decrease" if there is a rise above the normal.

- This kind of ordering for the opposite is "negative feedback", to restore the normal state. Most feedback mechanisms in our body are of this negative type.
- **Example:** TSH level.

- **Positive feed-back mechanisms** are very few. One example is that of uterine contractions during child-birth. Normal state of uterus is uncontracted, one contraction instead of commanding to come to normal gives a message to continue to contract further (positive feed-back till delivery is completed).

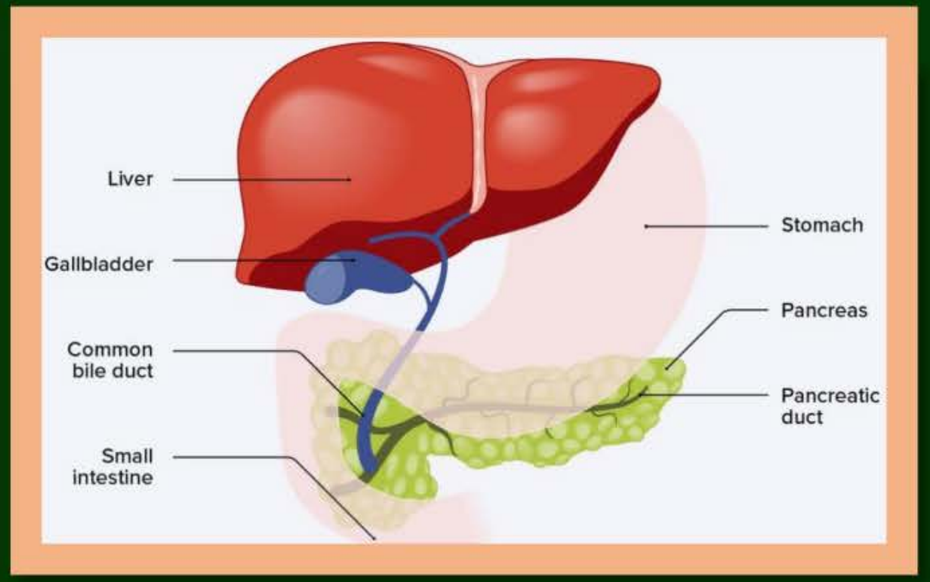
Negative Feedback Mechanism with reference to the TSH level in blood

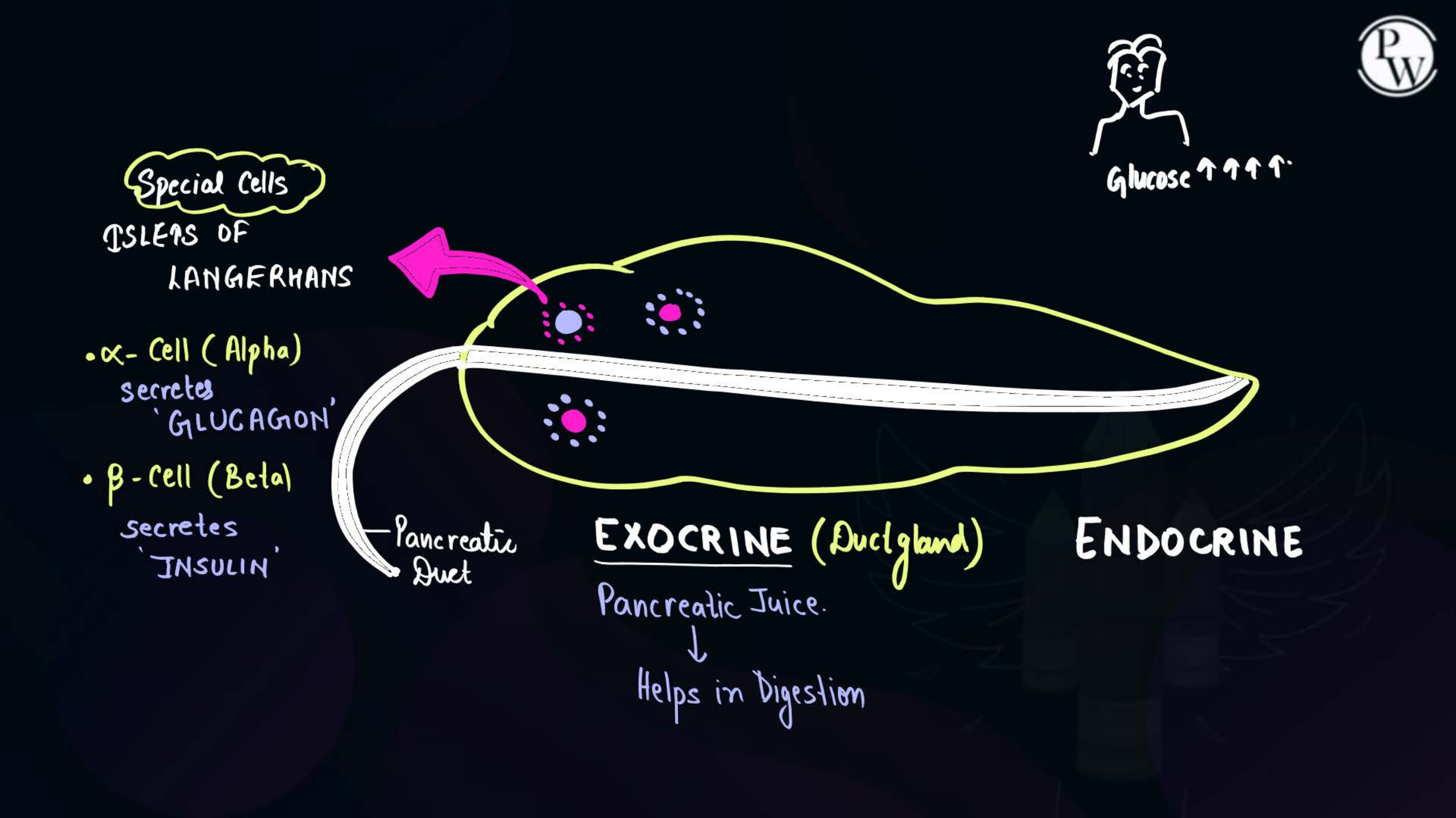


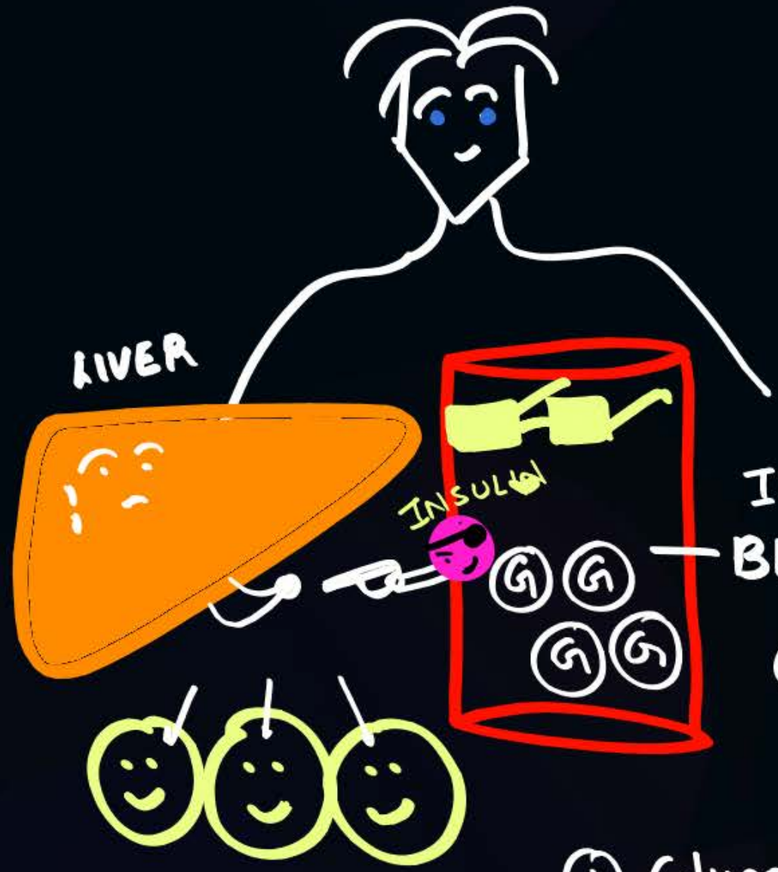
- When the level of thyroxine concentration exceeds the normal level in blood, then it inhibits the secretion of TRH by hypothalamus and the secretion of TSH from the anterior pituitary gland.

Pancreas

- Pancreas is both a duct gland as well as a ductless gland. (Mixed Gland, Heterocrine)
- As a duct gland, its secretion (pancreatic juice) is poured into the duodenum for digestion.
- As a ductless gland, it has special groups of hormone-secreting cells called Islets of Langerhans, which are scattered in the entire gland (islets : little islands).




Special Cells
ISLETS OF LANGERHANS
• α- Cell (Alpha) secretes 'GLUCAGON'
• β- Cell (Beta) secretes 'INSULIN'
Glucose ↑↑↑↑
Pancreatic Duct
EXOCRINE (Ductgland)
Pancreatic Juice.
↓
Helps in Digestion
ENDOCRINE



① Glucose enter in Body Cells
(Respiration)

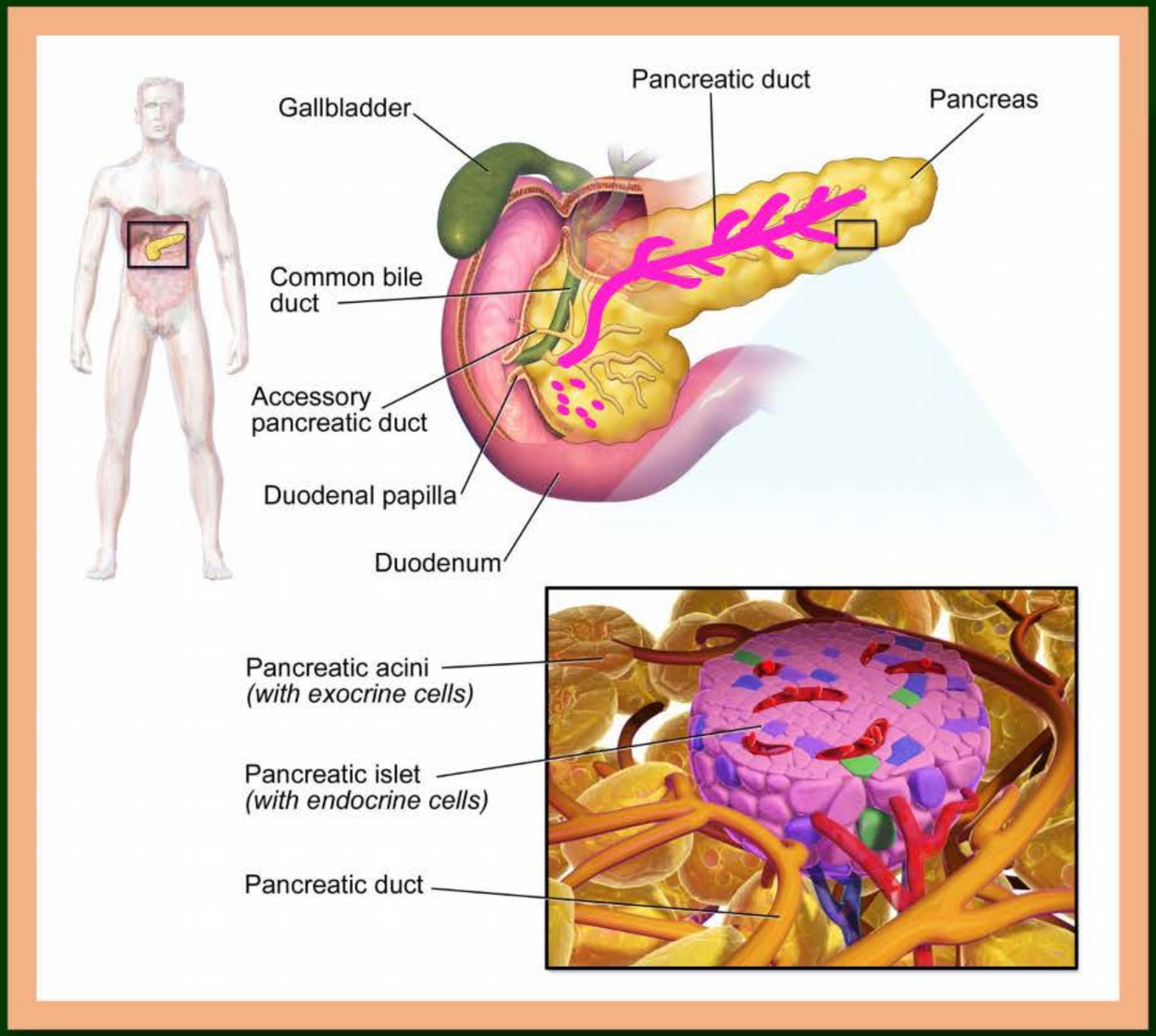
Glucose $+ O_2 \longrightarrow$ ATP

② Glucose will polymerise to form Glycogen [G+G+G+G]
Glycogenesis

When Blood Glucose is less, Pancreas will secrete GLUCAGON

Glucagon will break Glycogen to form Glucose in blood.

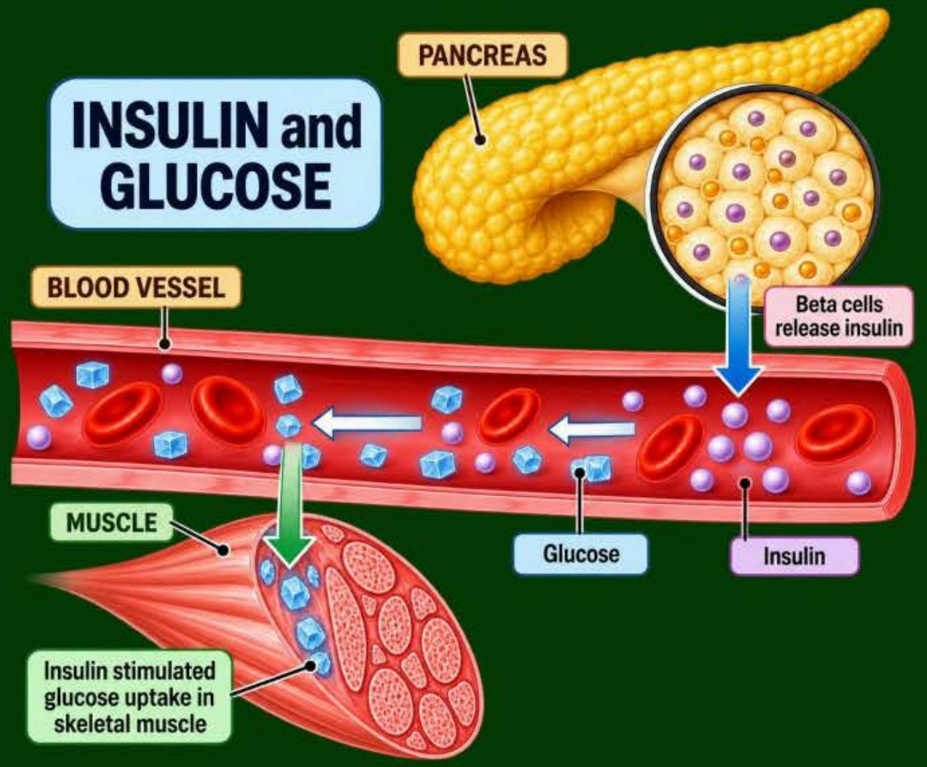
Glycogenolysis
break


Gallbladder
Pancreatic duct
Pancreas
Common bile duct
Accessory pancreatic duct
Duodenal papilla
Duodenum
Pancreatic acini
(with exocrine cells)
Pancreatic islet
(with endocrine cells)
Pancreatic duct

- **Insulin** secreted by beta cells checks rise of sugar level in blood. This it does in two principal ways:

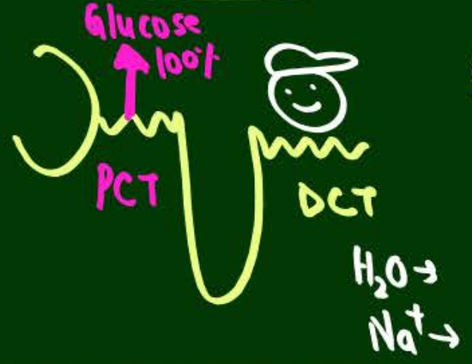
(i) It promotes **glucose utilisation** by the body cells, thereby reducing the blood sugar level.

(ii) It stimulates **deposition of extra glucose** of the blood as glycogen in liver and muscles.



Under secretion of Insulin

- Insufficient secretion of insulin causes diabetes mellitus or *hyperglycaemia*).
- The word "*mellitus*" means honey, referring to the passage of sugar (glucose) in urine.

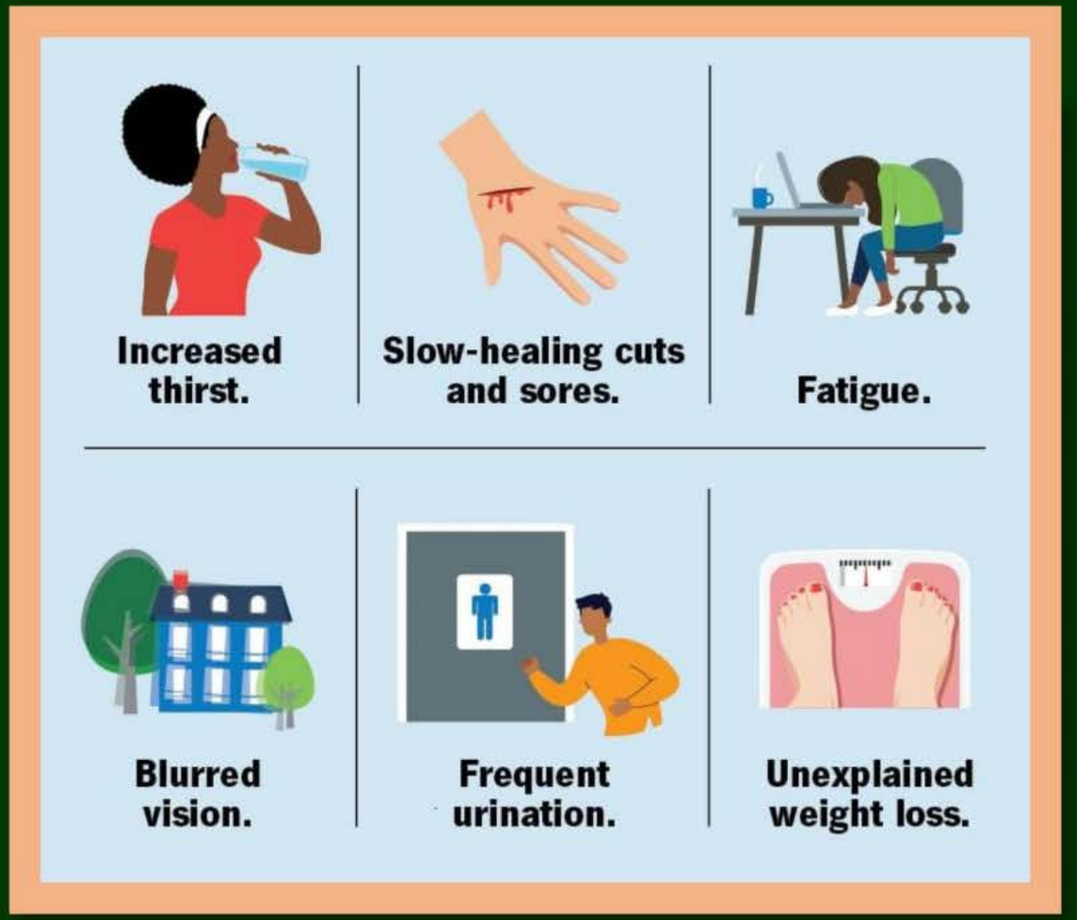


New Pattern

Excretory
ADH
URINE

Endocrine
ADH

Diabetes mellitus (More Glucose in Blood)



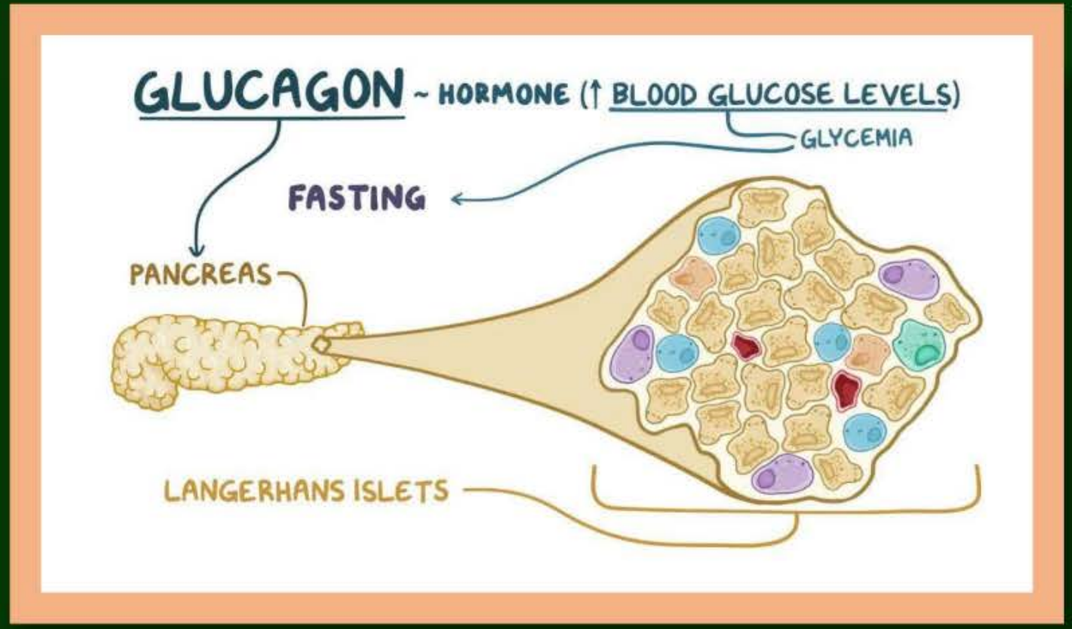
❑ **A diabetic person :**

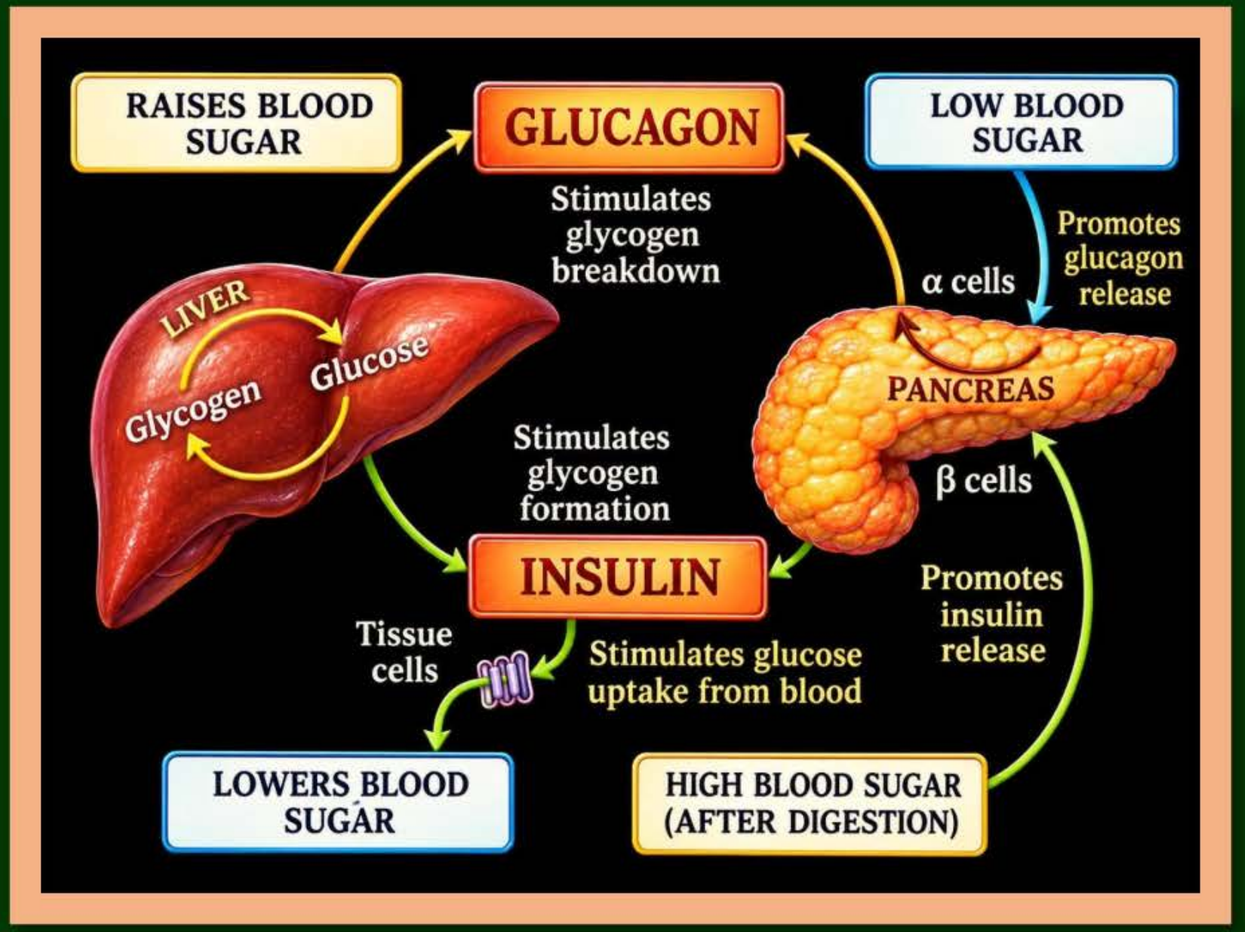
- Has high concentration of sugar in blood [hyperglycaemia - hyper : excess, glyce : sugar (glucose), emia : blood].
- Excretes a great deal of urine loaded with sugar.
- Feels thirsty because of the loss of water through too much urination.
- Loses weight and becomes weaker and weaker. In certain cases, the person loses eye-sight or vision.

Over secretion of Insulin

- Sugar level in the blood is lowered (hypoglycaemia, *hypo* : below).
- Brain may enter a state of coma if the level becomes too low even for a few minutes.

- Glucagon is secreted from alpha cells. It stimulates the breakdown of glycogen in the liver to glucose, thus it raises sugar level in the blood.

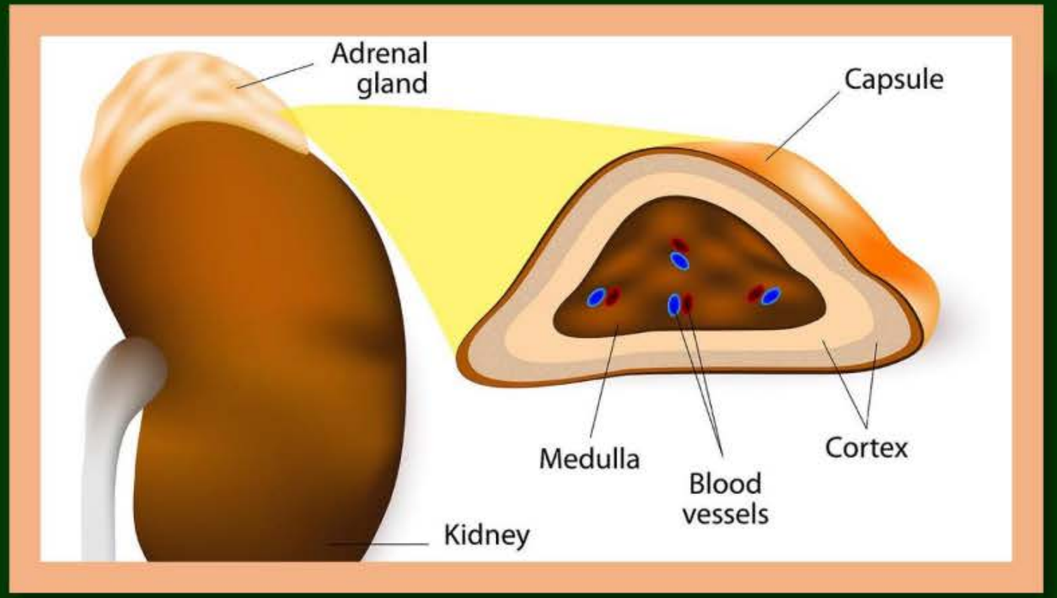


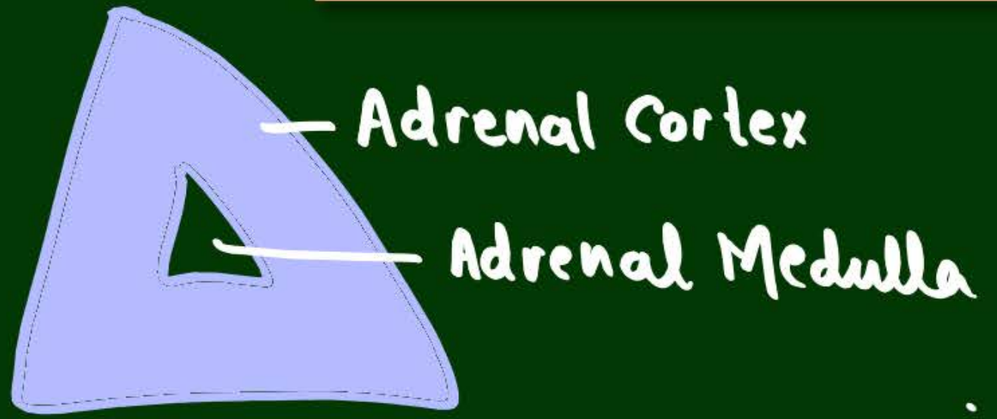


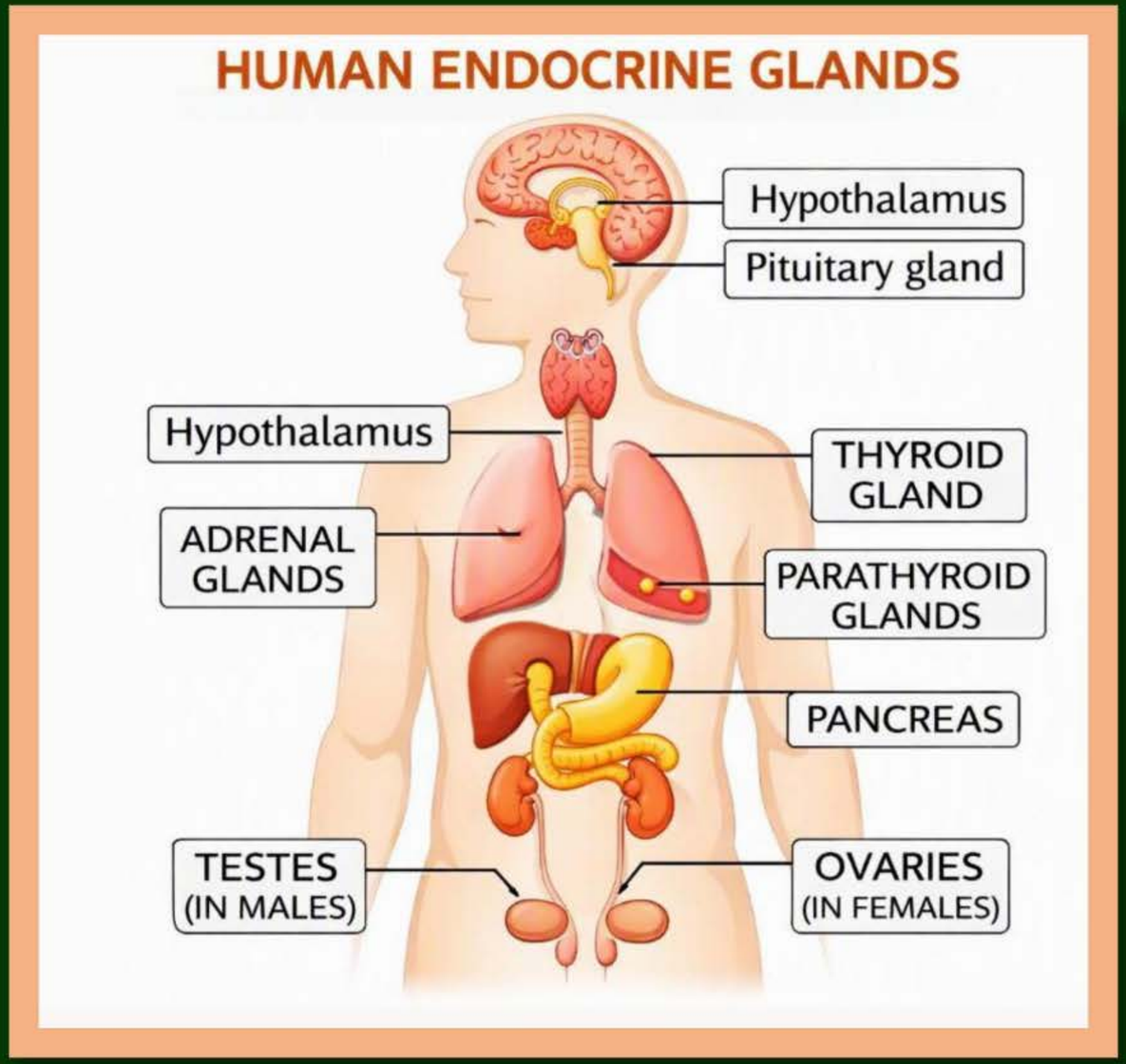
The role of pancreatic hormones (insulin and glucagon) in regulating blood sugar (glucose) level

Adrenal Glands

- The adrenal glands are like caps on the top of each kidney (ad: near, renal: kidney). Each Adrenal gland consists of two parts – adrenal medulla and adrenal cortex.

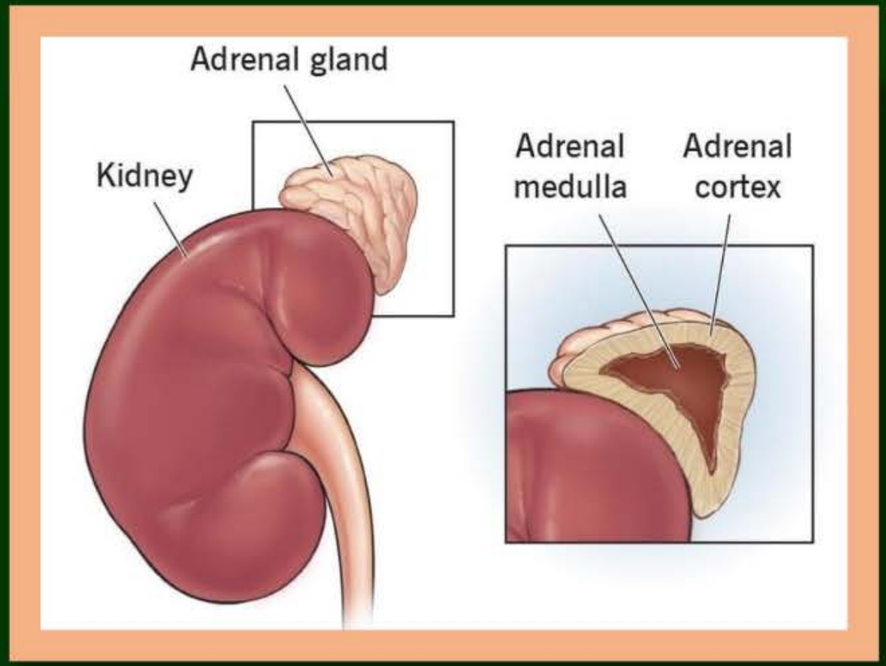




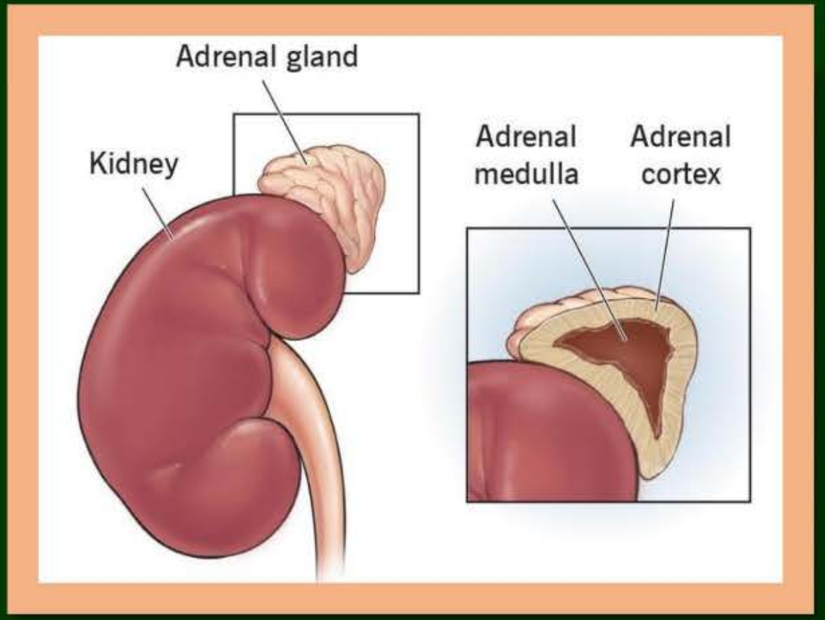

HUMAN ENDOCRINE GLANDS
Hypothalamus
Pituitary gland
Hypothalamus
THYROID GLAND
ADRENAL GLANDS
PARATHYROID GLANDS
PANCREAS
TESTES (IN MALES)
OVARIES (IN FEMALES)

(i) Adrenal medulla secretes adrenaline (also called epinephrine and noradrenaline). Sympathetic N.S

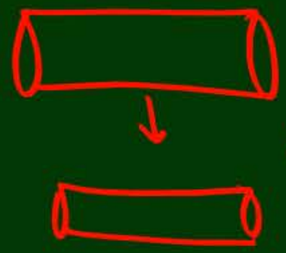
- It prepares the body to meet any emergency situation, for "fight" i.e. to face danger or for "flight", to run away from it.



- Extra hormone is released into the blood at the time of emotional stress.
- When excited or angry, our adrenals produce a lot of adrenaline.
- The gland itself is stimulated by the nerve endings of the autonomic nervous system.

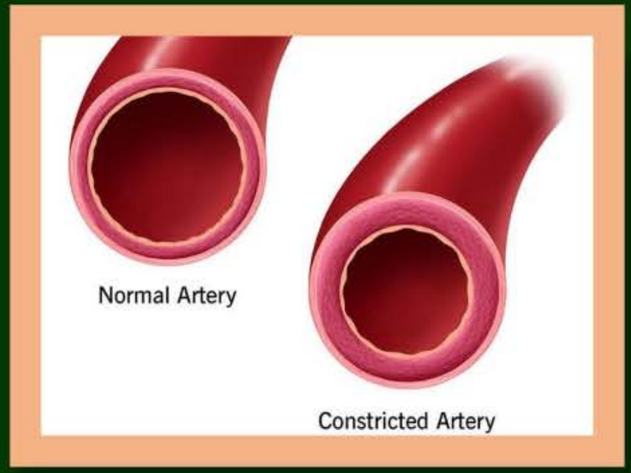


S. No.	Body parts	Effects of adrenaline	Biological advantage	Effects or sensation
1	Heart	Beats faster Blood pressure increases ↑	Sends more glucose and oxygen to the muscles	Thumping heart 



Vasoconstriction
Constrict the Blood Vessel → except CORONARY ARTERY ↳ (Heart)

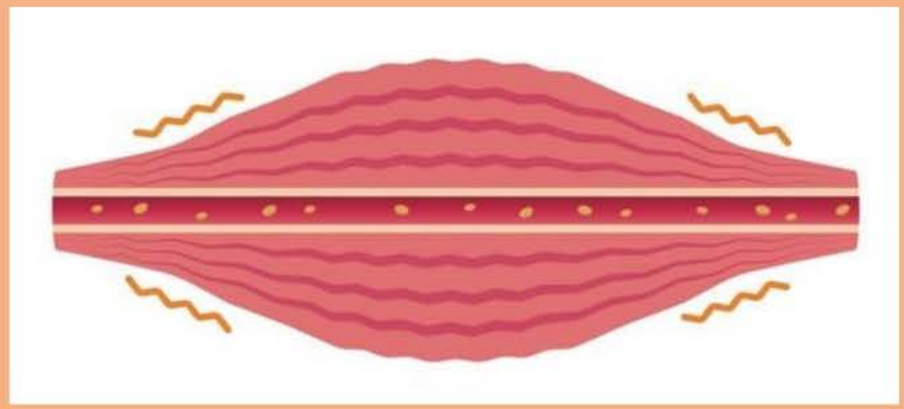
S. No.	Body parts	Effects of adrenaline	Biological advantage	Effects or sensation
2	Breathing centre of the brain	Faster and deeper breathing	Increased oxygenation of the blood; rapid removal of carbon dioxide	Panting

S. No.	Body parts	Effects of adrenaline	Biological advantage	Effects or sensation
3	Arterioles of the skin	Constricts them	Less blood going to the skin means more is available to the muscles	Person turns pale 

S. No.	Body parts	Effects of adrenaline	Biological advantage	Effects or sensation
4	Arterioles of the digestive system	Constricts them	Less blood to the digestive system, allows more blood to reach the muscles.	Dry mouth

S. No.	Body parts	Effects of adrenaline	Biological advantage	Effects or sensation
5	Muscles of body	Tenses them	Ready for immediate action	Tense feeling; shivering





S. No.	Body parts	Effects of adrenaline	Biological advantage	Effects or sensation
6	Liver	Conversion of glycogen to glucose	Glucose available in blood for energy production	No sensation

S. No.	Body parts	Effects of adrenaline	Biological advantage	Effects or sensation
7	Fat deposit	Conversion of fats into fatty acids	Fatty acids available in blood for muscle contraction	No sensation

- In addition, adrenaline also increases clotting capacity of the blood, dilates pupil of the eye, stimulates uterine contraction during labour (delivery of baby), stimulates muscle of the hair follicles. (These functions are in general the same as those of the sympathetic autonomic nervous system).

QUESTION

A gland which secretes enzymes and hormones both is

A. Adrenal

B. Pancreas

C. Thyroid

D. Pituitary

QUESTION

Insulin is secreted by

A Beta cells of pancreas

B delta cells of pancreas

C Alpha cells of pancreas

D gama cells of pancreas

QUESTION

Islets of Langerhans are located in the

A Kidney

B Liver

C Throat

D Pancreas ✓

QUESTION

Exophthalmic goitre is caused due to overactivity of

A Thymus

B Thyroid

C Parathyroid

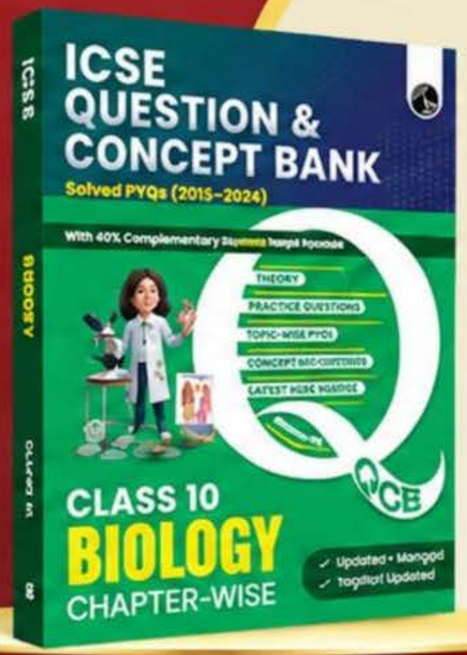
D Adrenal cortex

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